

A PCB design approach for best EMC

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Introduction

- Tim Jarvis BSc CEng MIEE MIEEE
 - RF /Analogue Engineer
 - Since graduating in 1983
 - Offshore and Naval Sonar
 - DSP /Embedded
 - Detour 1987 to 1997
 - Telecommunications industry
 - RF & EMC since 1991
 - Founded RadioCAD in 2000 following redundancy.

■ Presentation History

- Approval 1999 Conference
 - Best practice in circuit board design
- IEE New Issues in Design April 2004
 - Practical Power Planes for EMC
- Automotive EMC 2004
 - Practical Power Planes for EMC.



Design Approaches

- Ignore EMC requirements
- Ad-hoc (random) design rules
- Standard design rules

**first principles
systematic
approach**



Pre-PCB Questions

- What is our EMC target?
 - Emissions
 - Immunity
 - ESD, etc.
 - Market access
 - Product safety
 - Other directives
 - Conflicting standards



Pre-PCB Questions (continued)

- What are our product constraints?
 - Anticipated volumes
 - High volume = spent time engineering for lowest cost = minimum PCB layers (2-layers)
 - Low volume = minimise engineering time = more PCB layers, plenty of component options.
 - Component technologies
 - Fast semiconductors (faster than we need?)
 - Cheap passives (spend £s to save pence?)



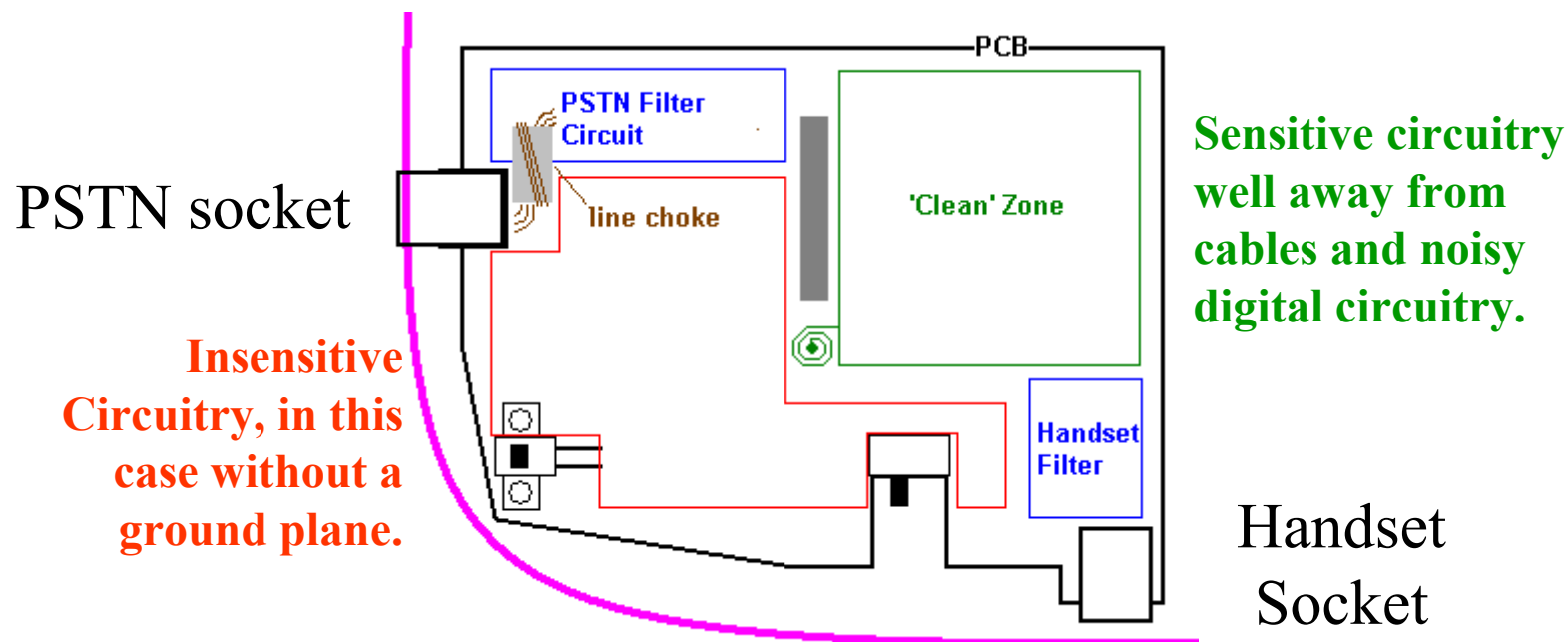
Pre-PCB Questions (continued)

- Segregation at the schematic level
 - What are our system blocks?
 - Which blocks are noisy?
 - Which blocks are sensitive?
 - What about ports and connectors?
 - Import /export EMI /ESD
 - Interact with shield /enclosure boundary
 - Buy or build EMI filters?



PCB Step 1 - Placement

- RULE1 - Never use the auto-placer and always segregate sub-systems



In this real-world example the noisy digital circuitry was placed on another PCB.



Filter Placement

- RULE2 - Always place filters on system /sub-system boundaries **EXAMPLE**

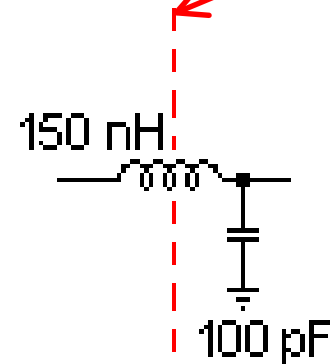
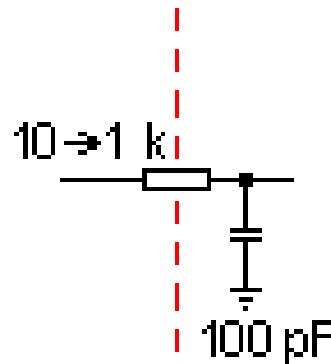
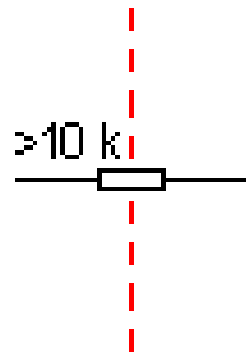


Lay the inline component across the boundary with shunt component(s) immediately behind it.



Simple DIY Filters

■ Asymmetric Filters



Why not show
the sub-system
boundary on
your schematic?

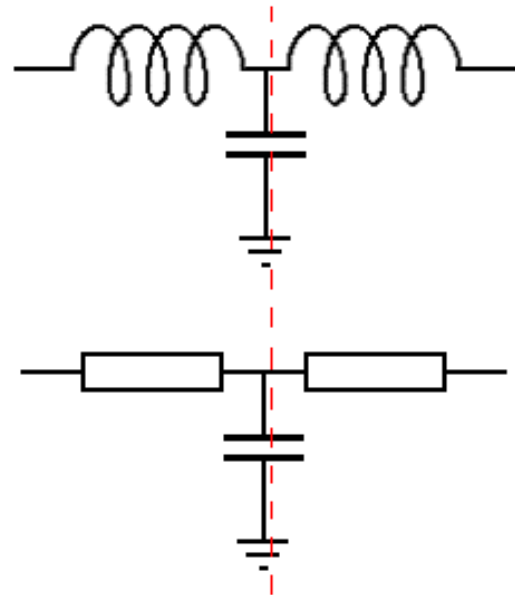
High Impedance Signal	Medium Impedance Signal	Low Impedance Signal
Use existing resistance of 10 kΩ or greater.	Use existing resistance of 1 kΩ or greater and add 100pF capacitor.	Add 150 nH and 100pF capacitor.

Remember components limitations !
0603 NPO 100pF capacitor, SRF @ ~900 MHz
0805 150 nF chip inductor, SRF @ ~900 MHz

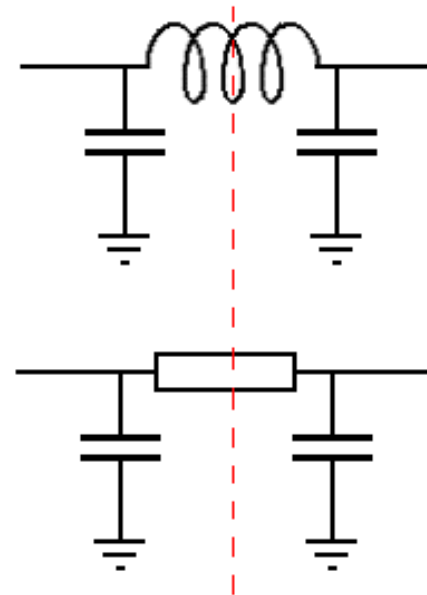


Simple DIY Filters (continued)

■ Symmetrical Filters



T-Filters



π -Filters

Always look at attenuation-v-frequency curves for components. Never rely on component values alone.

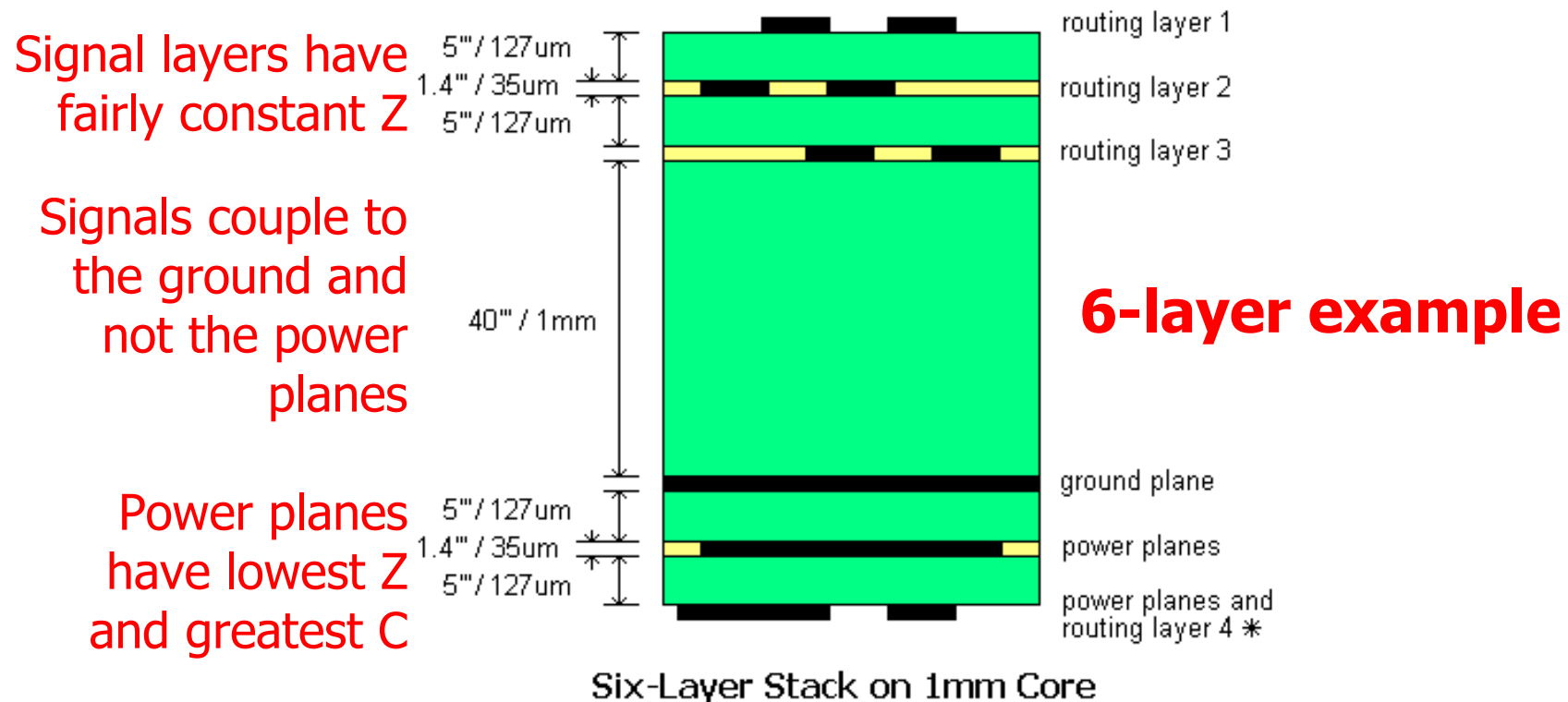


PCB Step 2 – Assign Layers

- Ground Plane?
 - Yes always, big and uncut
 - (only cut if you really know what you're doing)
- Power Planes, do we need them?
 - It depends who you talk to
 - 1 – look at the theory
 - 2 – look at the practice



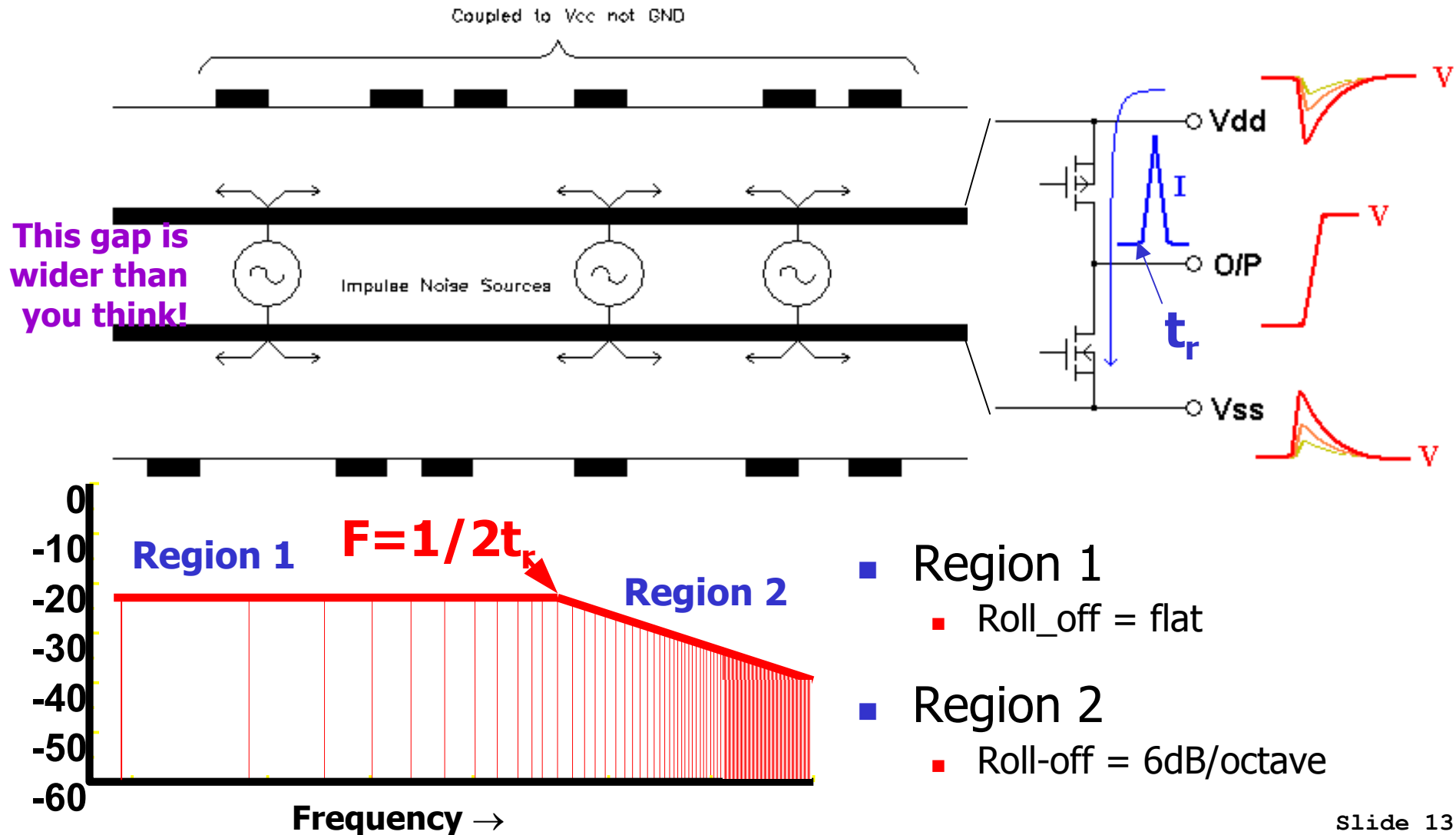
Standard multi-layer construction



**1 mm and 0.8mm cores are common.
= 0.13 and 0.23 mm prepreg**

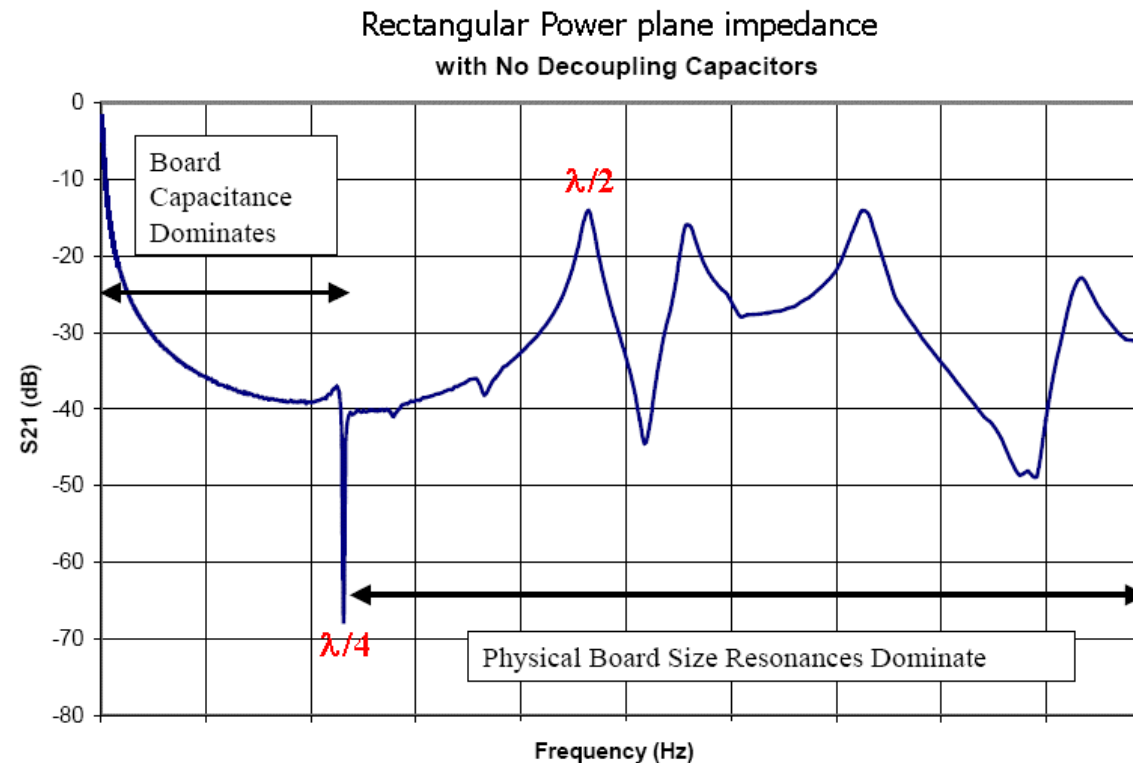


Avoid this stack order...





Power plane size

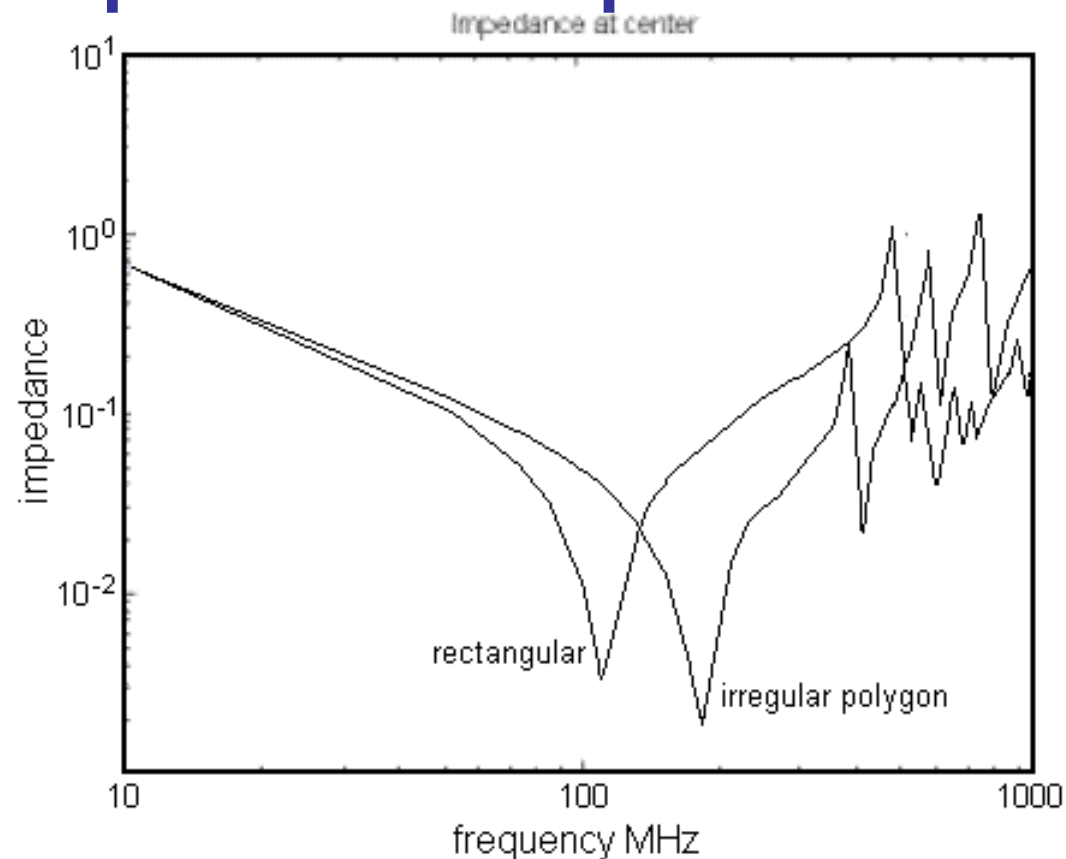


300x250mm PCB courtesy of Dr. Bruce Archambeault IBM

Small is best
(≤ 75 mm for 1 GHz)



Power plane shape

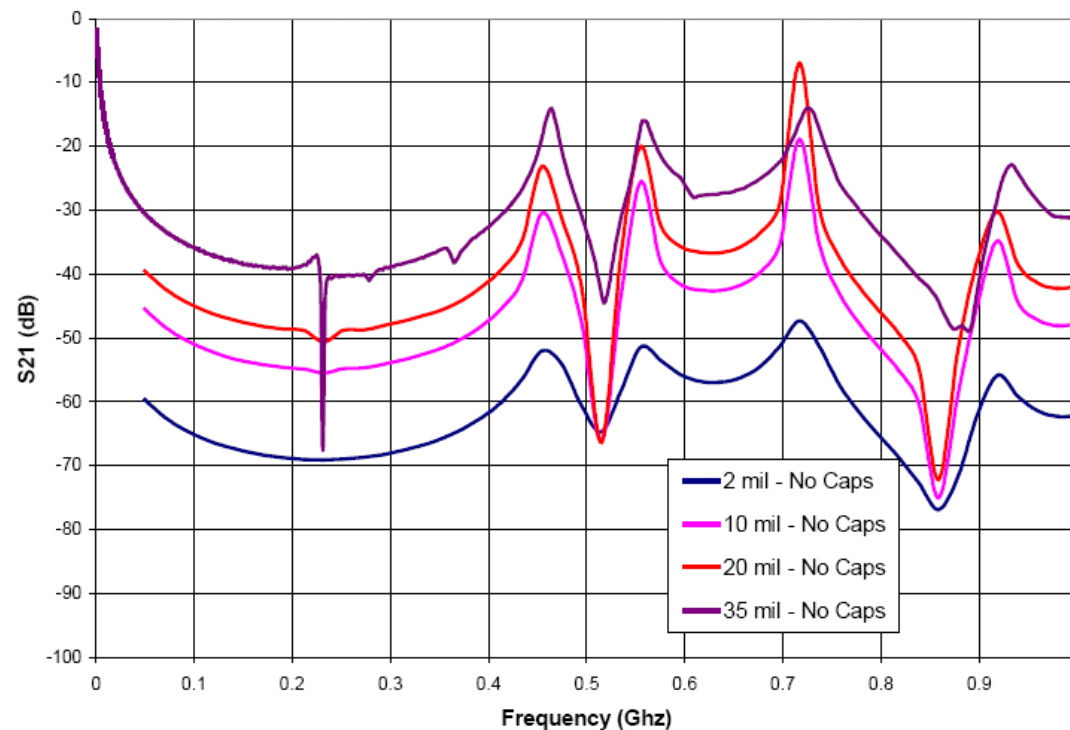


Non-trapezoidal irregular shapes are best
(but not much better)



Power Plane Separation

Rectangular Power Plane Impedance
with Various Power/Ground Plane Separation and No Capacitors



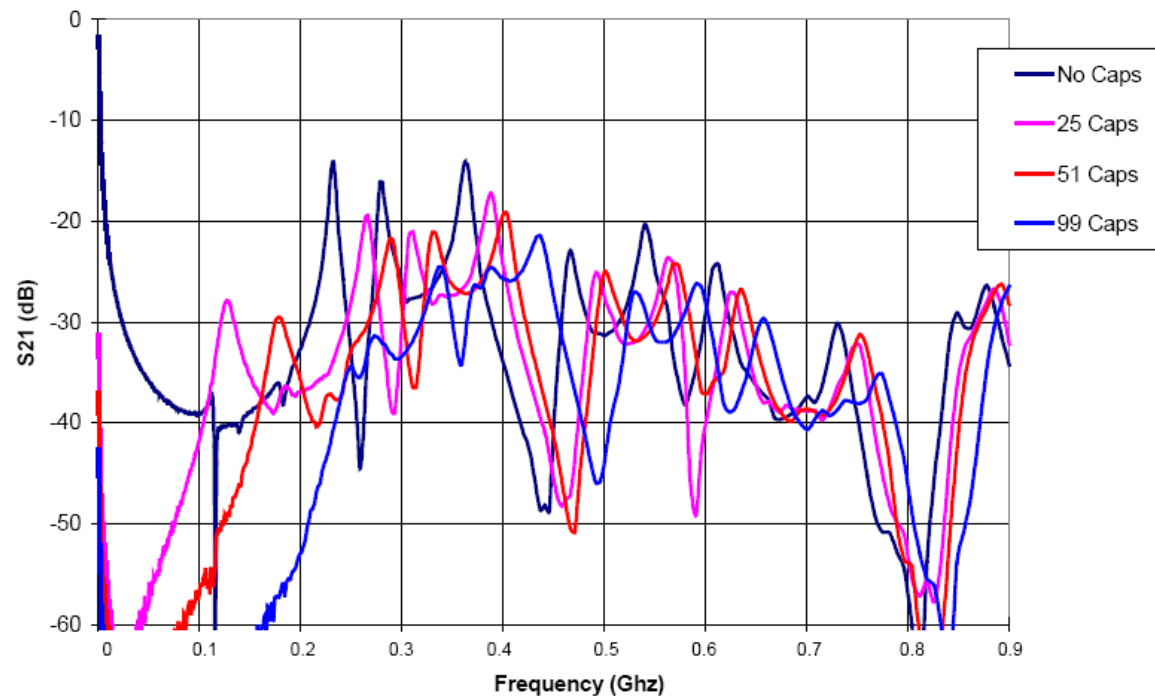
300x250mm PCB courtesy of Dr. Bruce Archambeault IBM

Close is best
($\leq 2''$ requires specialist techniques)



Interaction with decoupling caps

Rectangular Power Plane with 0805 decoupling caps at intervals
With Various Amounts of .01 uf Decoupling Capacitors



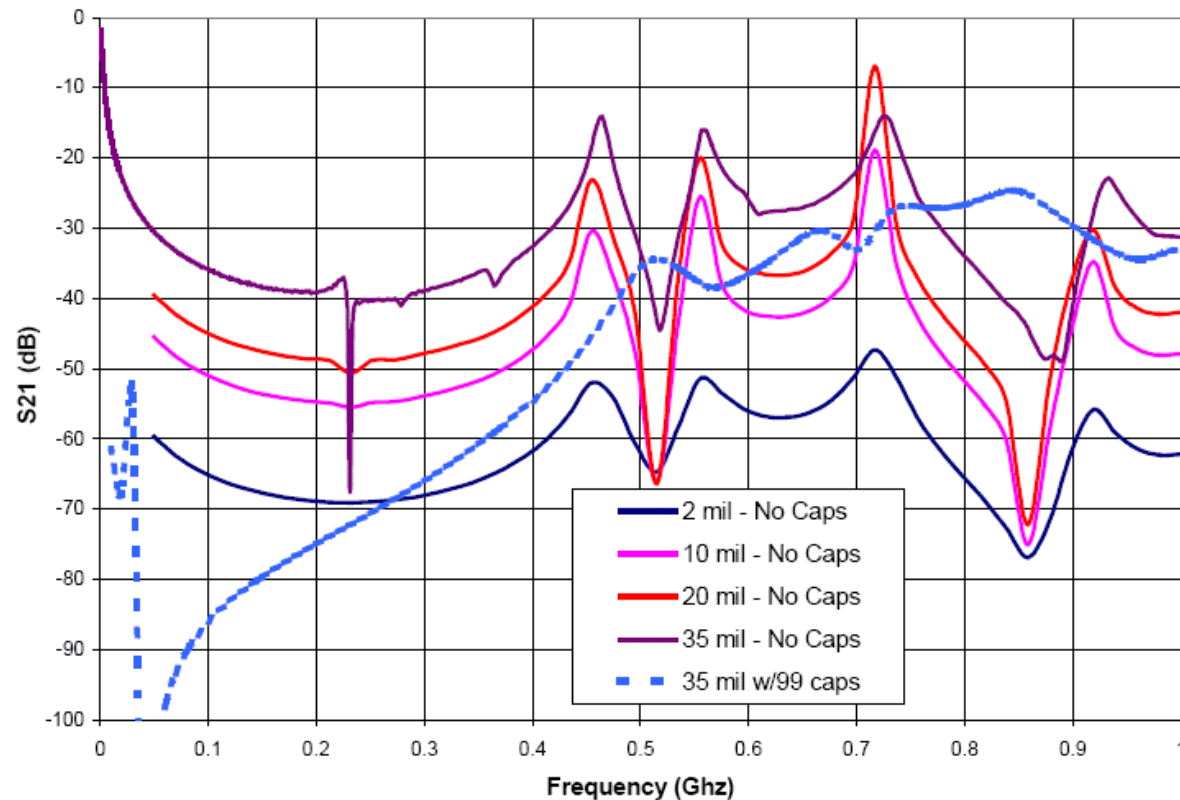
courtesy of Dr. Bruce Archambeault IBM

More caps the better
(but they have little effect above $\lambda/2$)



Interaction with decoupling caps (continued)

Rectangular Power Plane Impedance
with Various Power/Ground Plane Separation and No Capacitors



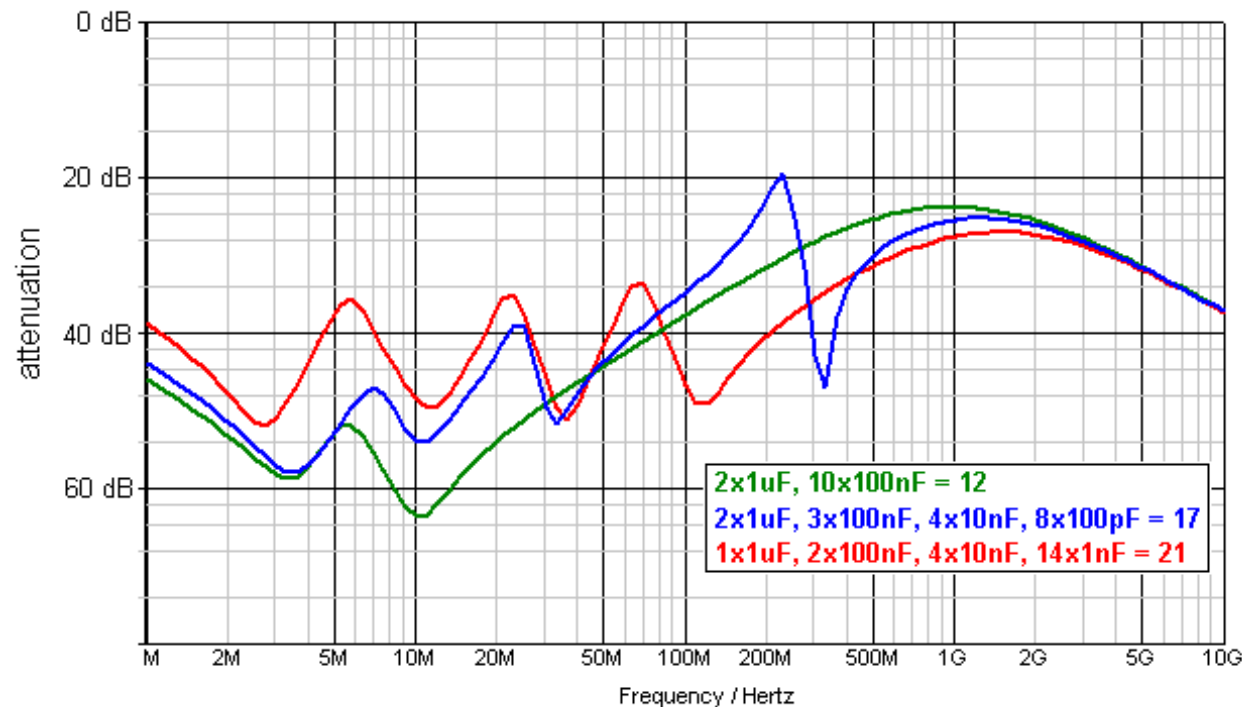
300x250mm PCB courtesy of Dr. Bruce Archambeault IBM

**2''' plane separation is better than 99 * 10nF Caps
with a 0.8mm separation @ freq>1/4**



What value of decoupling cap?

Comparison of decoupling capacitor arrays
(includes via and chip parasitics but excludes power plane effects)

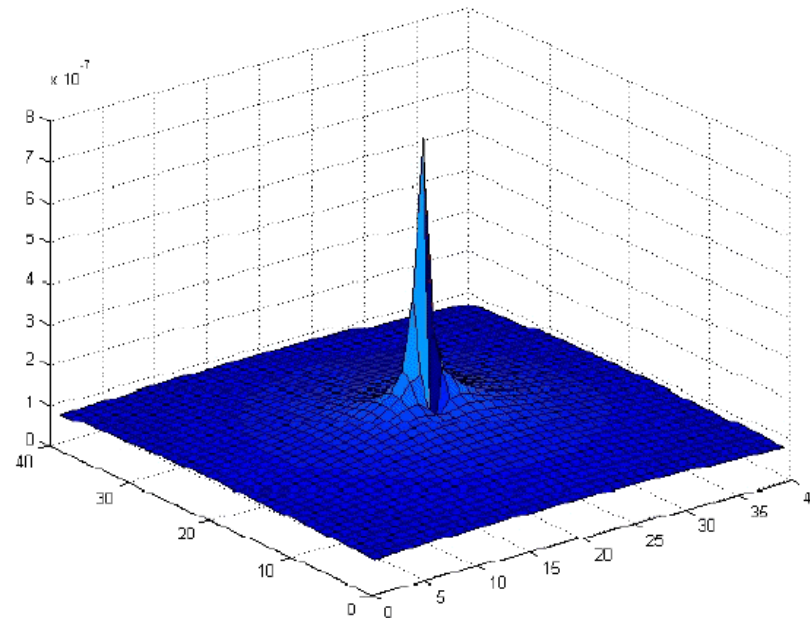


Plain old 100nF caps are usually best
(lots and lots of them)



Why we might want power planes

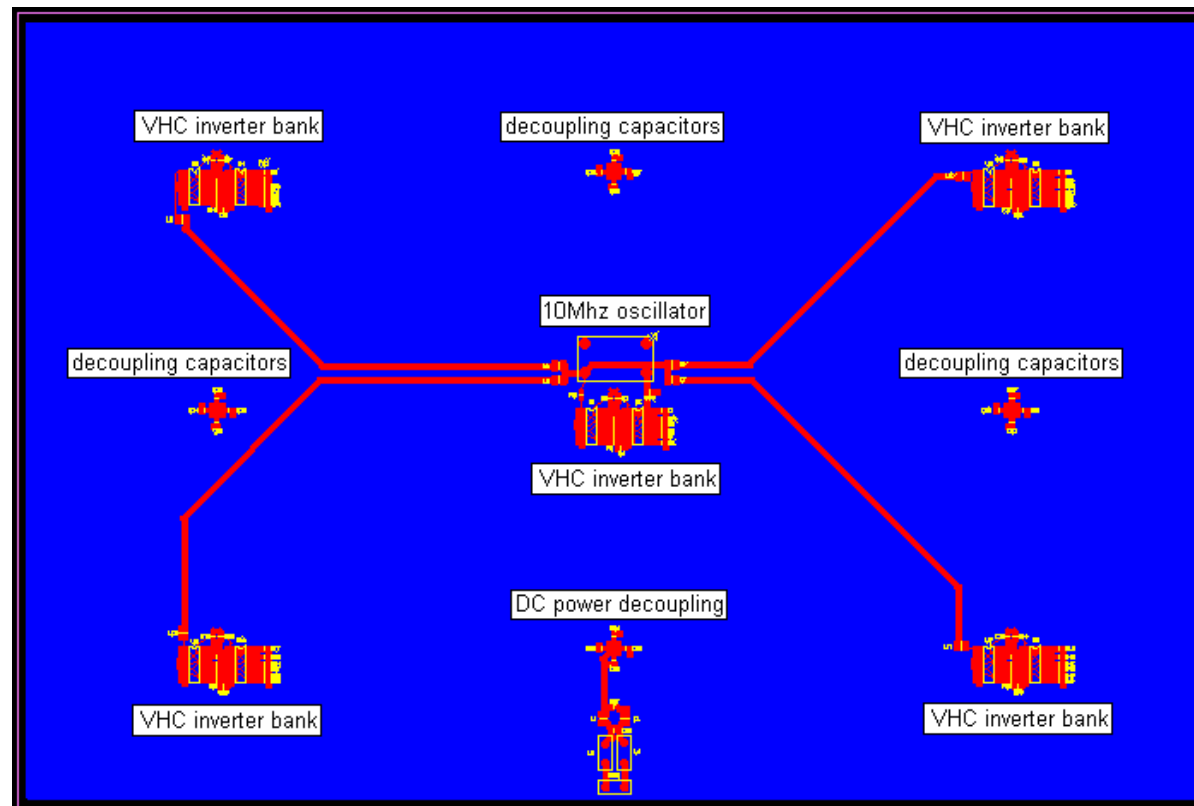
500 MHz via @40 mils from Source



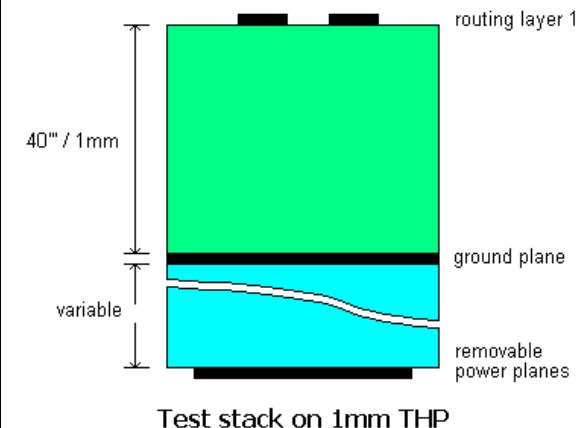
Current spreading in a power plane
reduces noise coupling between power pins



Power Plane Test JIG



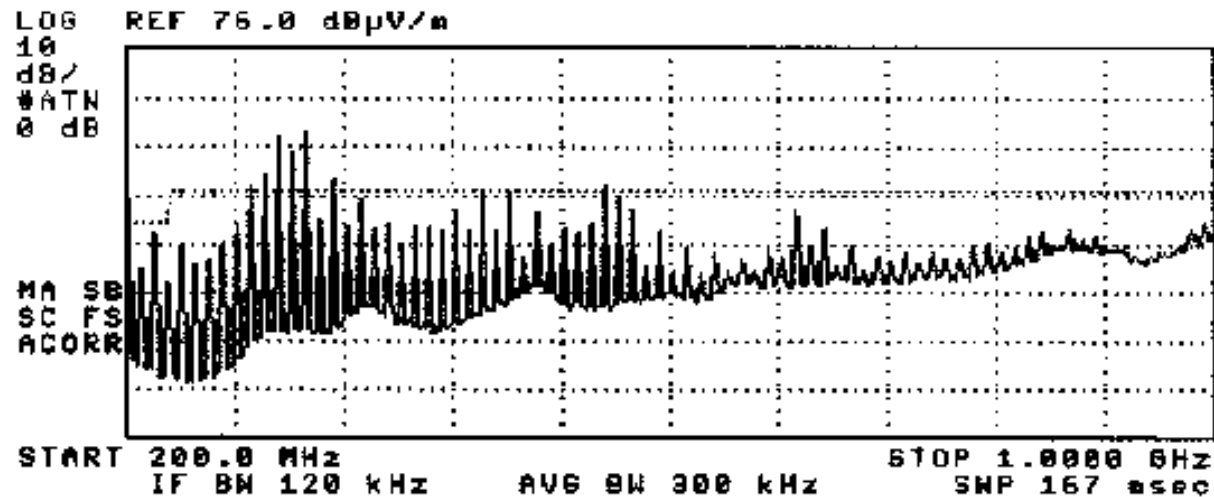
Ground plane and routing on 300x200x1mm THP FR4 PCB. Each decoupling capacitor has a 0.8mm via hole for stitching through to a removable power plane.



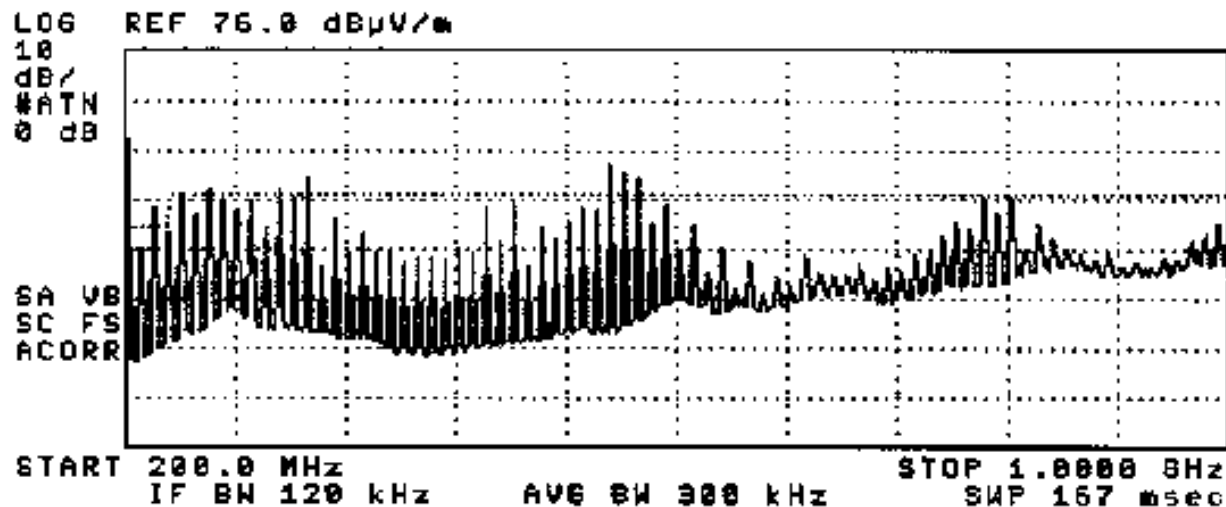
Has 5 evenly spaced emitters all connected to one 10 MHz oscillator via filtered traces. Has provision for additional decoupling caps.



300x200mm power plane @ 1.6mm, 10x100nF caps



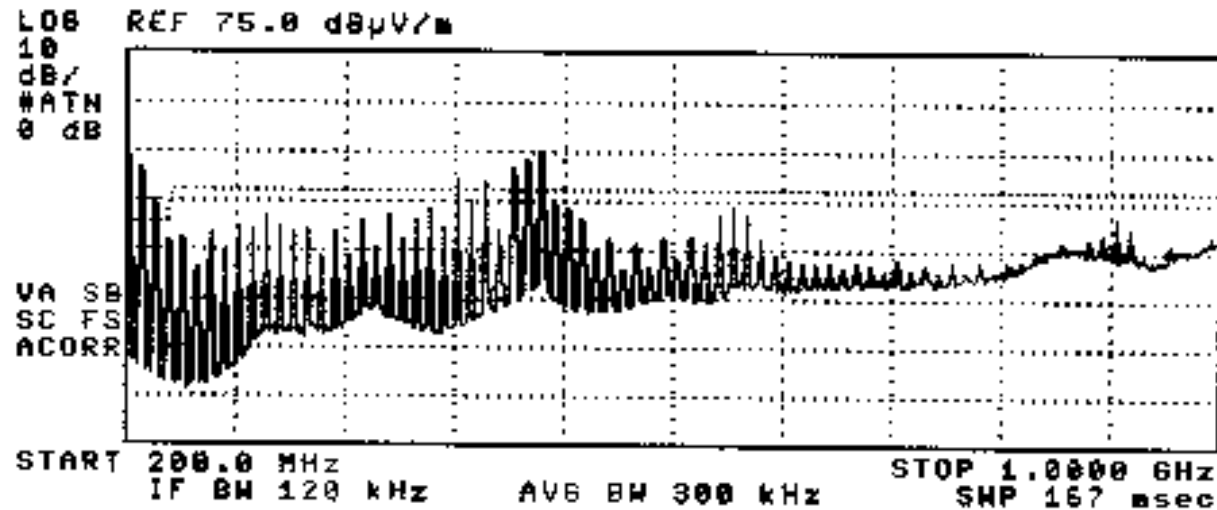
Horizontal



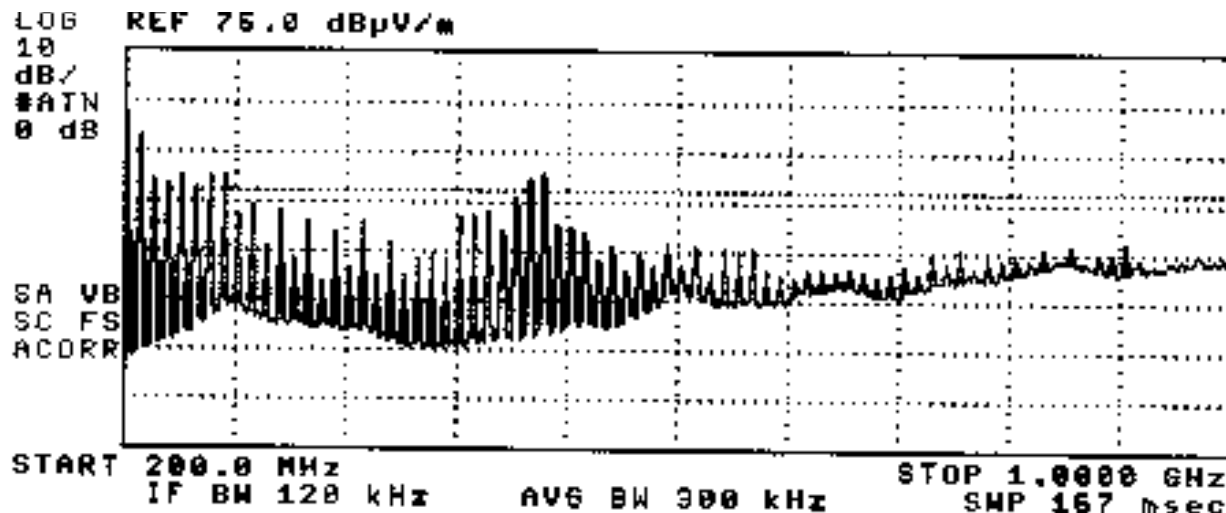
Vertical



300x200mm power plane @ 0.2mm, 10x100nF caps



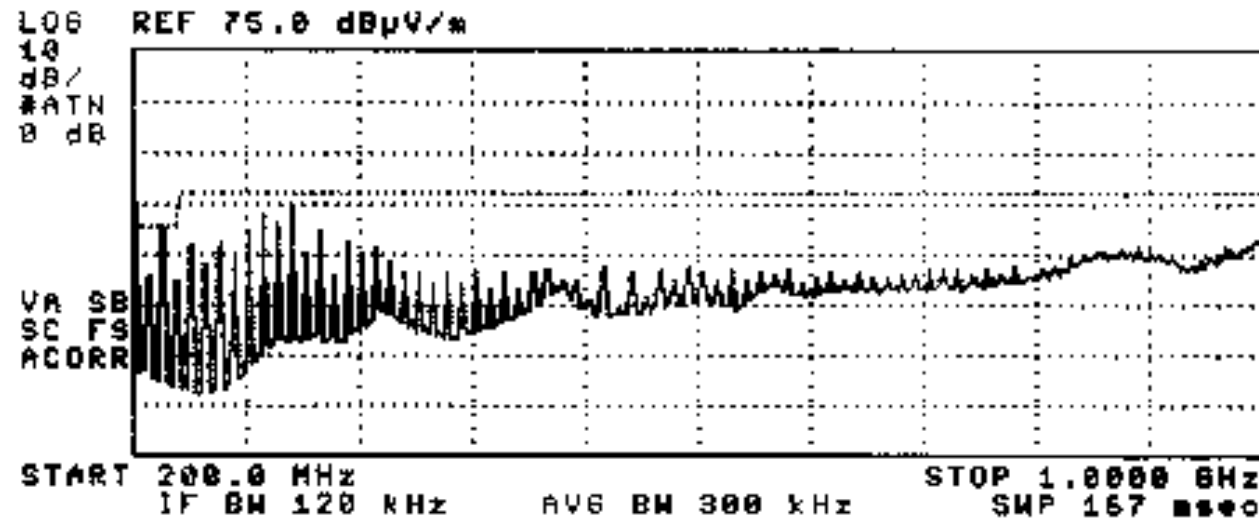
Horizontal



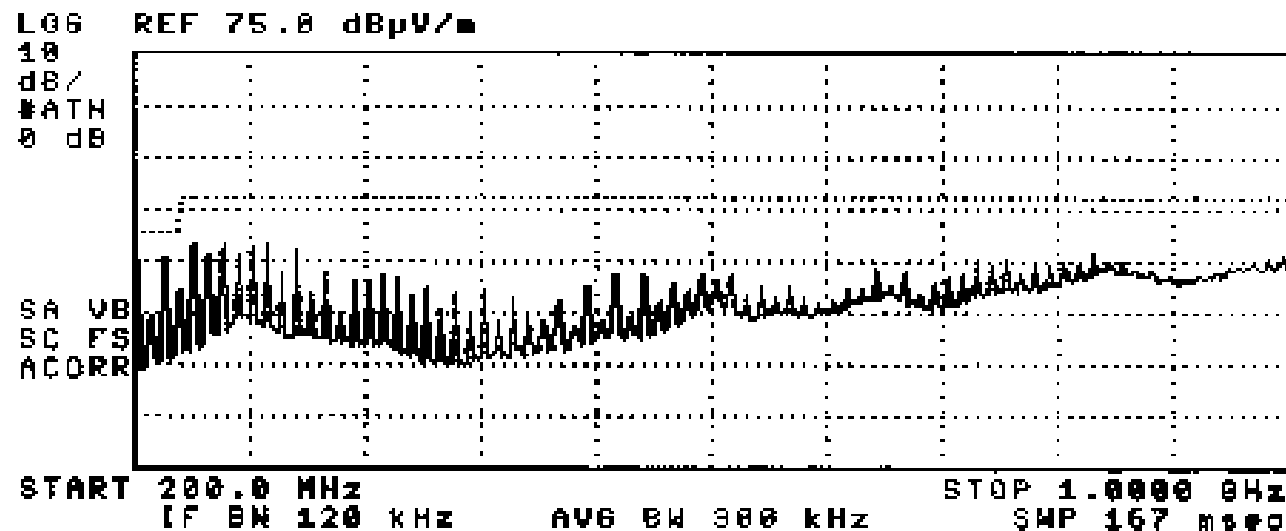
Vertical



300x200mm power plane @ 50 μ m, 35x100nF caps



Horizontal



Vertical



No power plane, 10x100nF caps!

LOG REF 75.0 dBμV/m

10

dB/

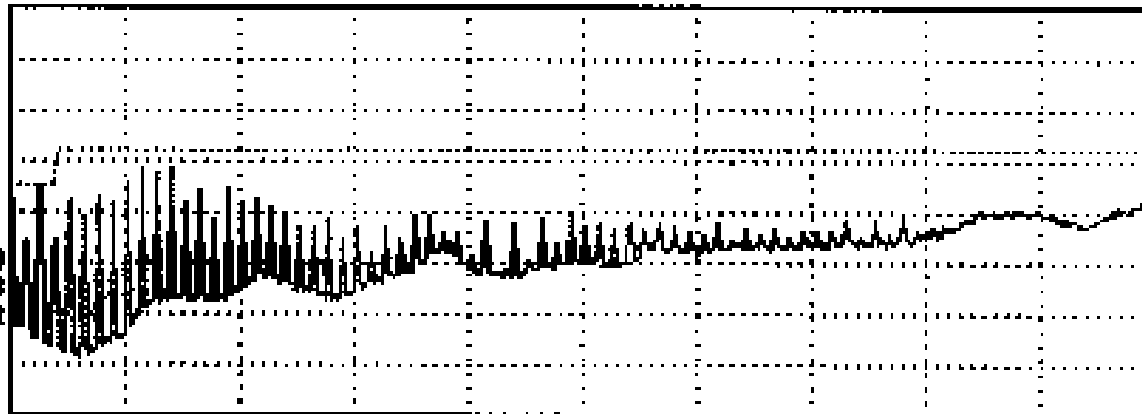
#ATTN

0 dB

VA SB

SC FS

ACORR



START 200.0 MHz

IF BW 120 kHz

AVG BW 900 kHz

STOP 1.0000 GHz

SWP 167 msec

LOG REF 75.0 dBμV/m

10

dB/

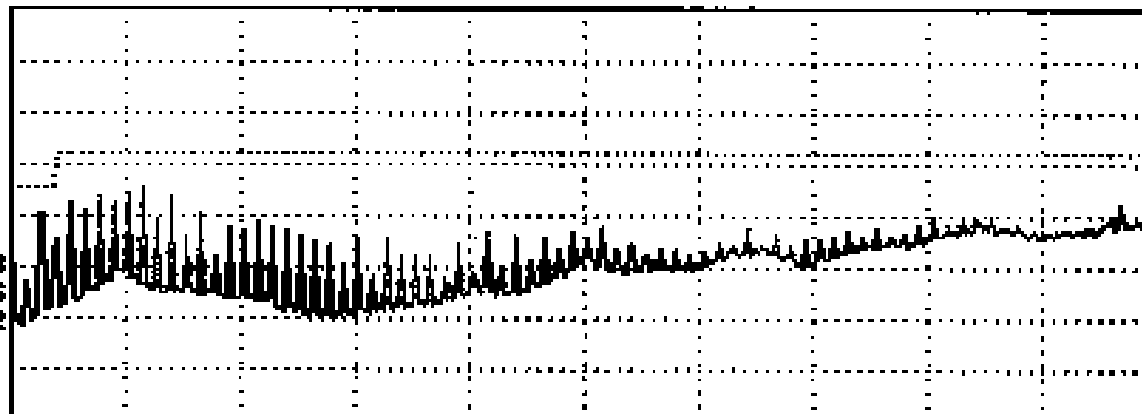
#ATTN

0 dB

SA VB

SC FS

ACORR



START 200.0 MHz

IF BW 120 kHz

AVG BW 900 kHz

STOP 1.0000 GHz

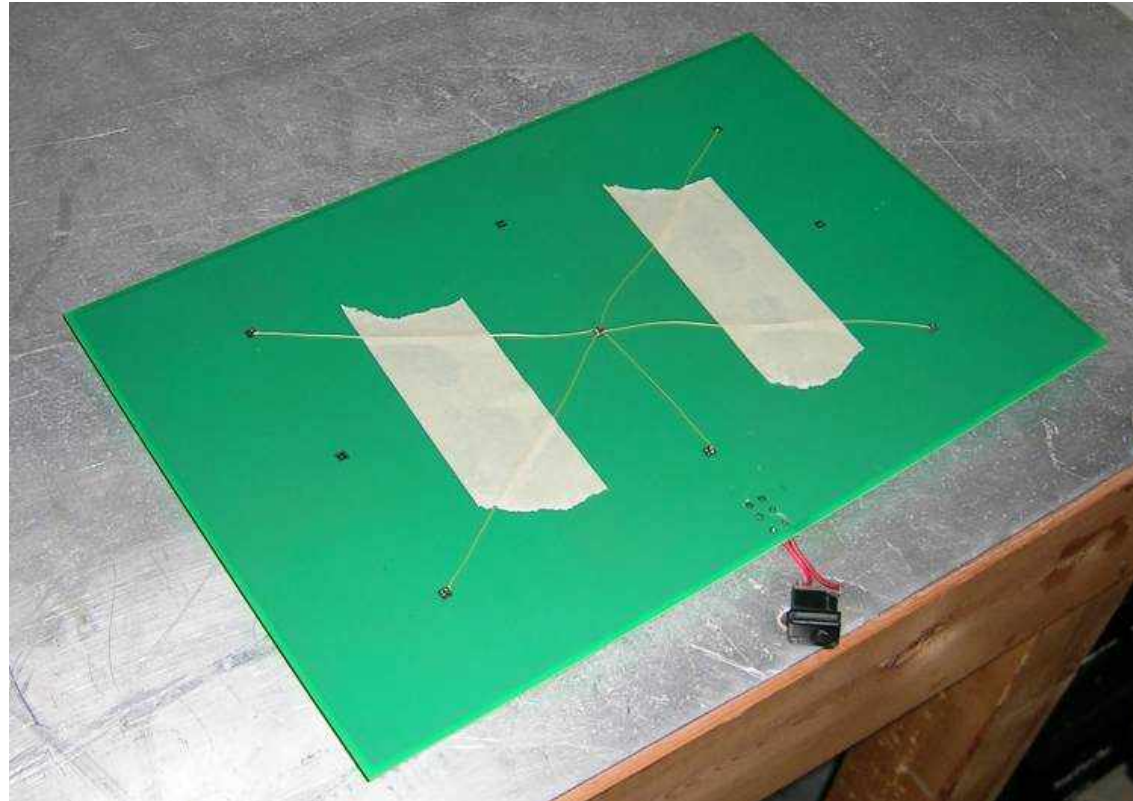
SWP 167 msec

Horizontal

Vertical



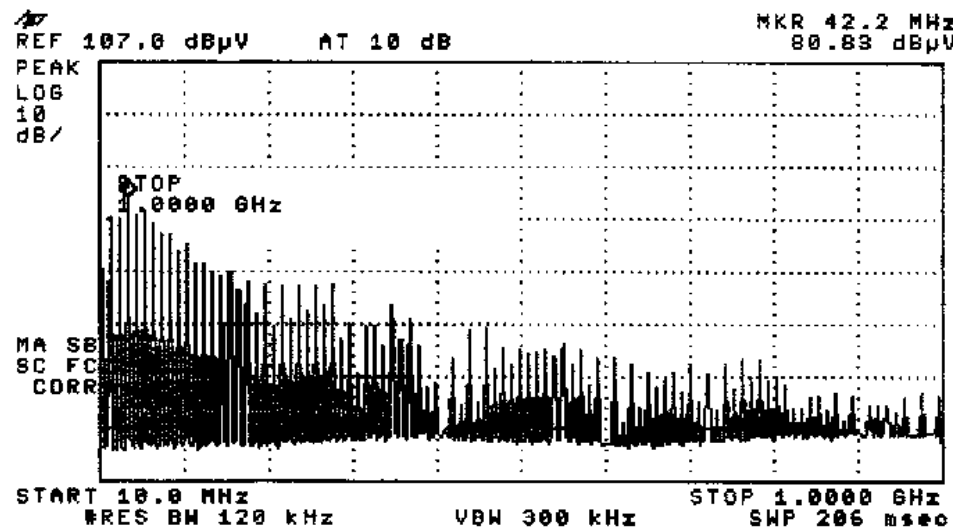
No power plane



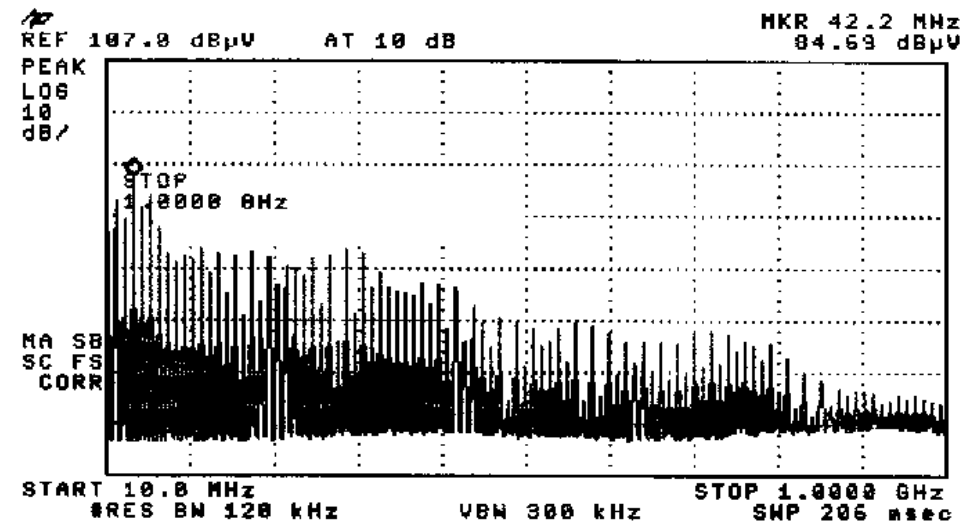
5 leads, $\phi 0.4\text{mm}$, close to ground plane



Comparison of Vcc Noise



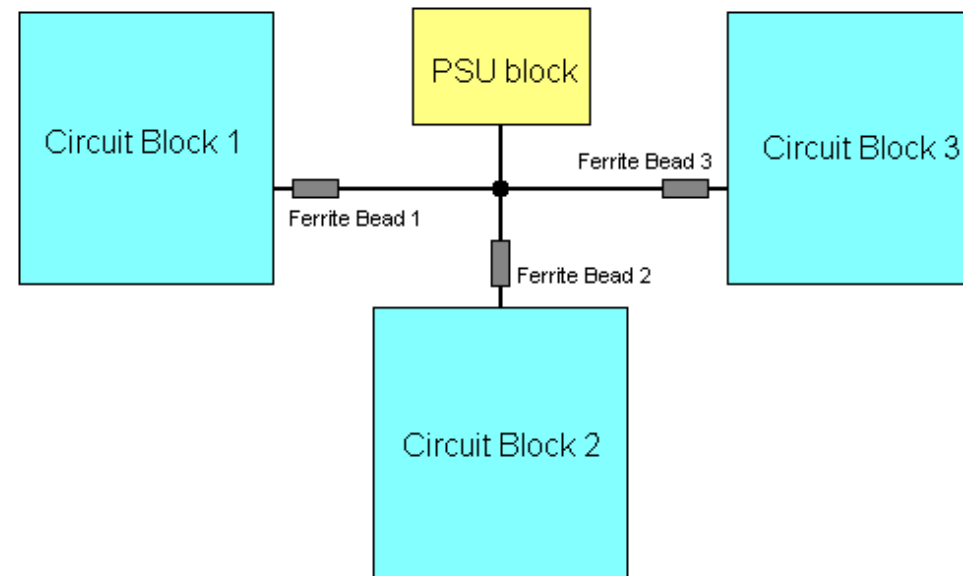
- 50µm power plane
 - 35 caps
 - Vcc noise ~12mV RMS



- No power plane
 - 10 caps
 - Vcc noise ~20mV RMS



Smaller 'regional' power planes



- Segment board into smaller regions
- Each has a regional power plane <75mm (1GHz)
- Each is interconnected via ferrites or inductors
- Each region has its own local decoupling caps



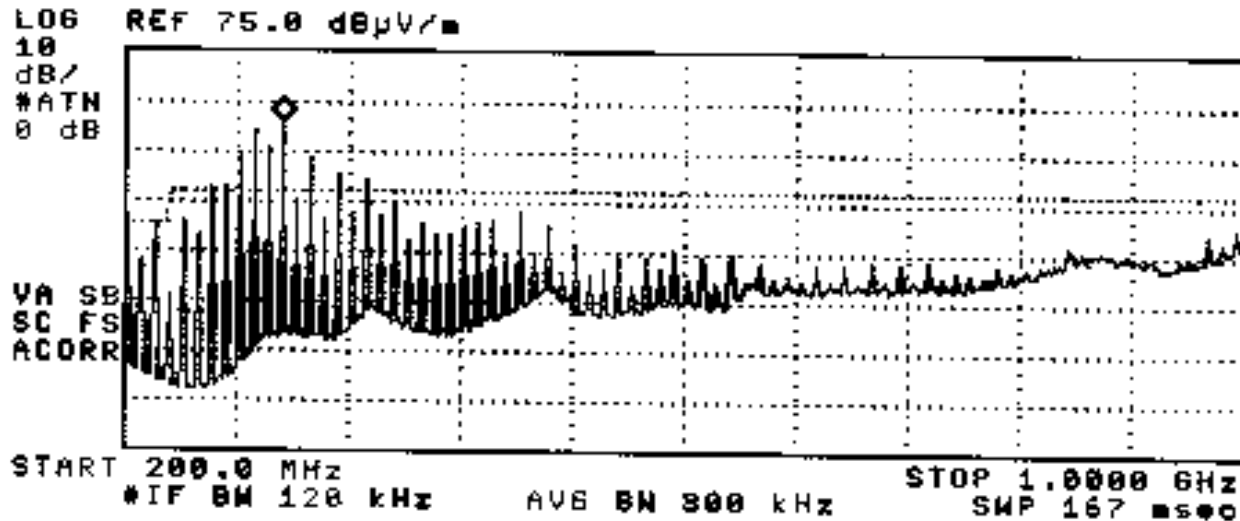
Example: 5 regional planes



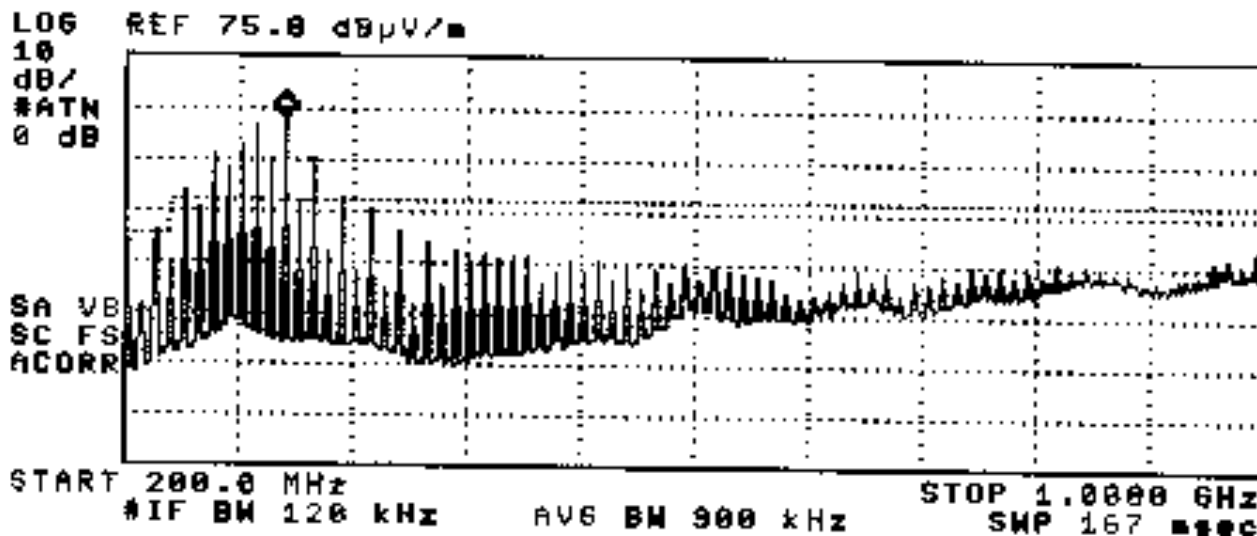
- Each region is connected via BLM18RK221SN1
- Each region has 4x100nF local decouplers
- A further 16x100nF caps decouple the remaining plane



75x75mm regional power planes @ 1.6mm, 4x100nF caps each



Horizontal



Vertical

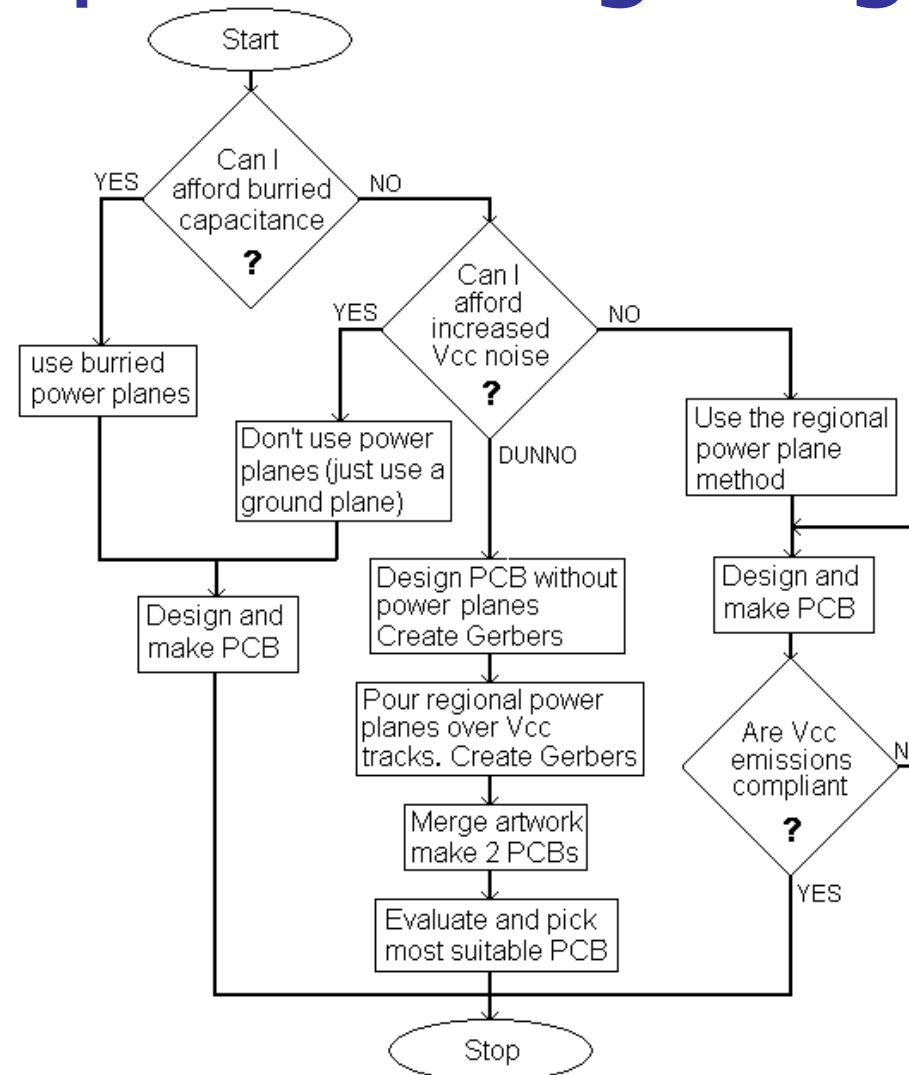


An interesting question...

- Why is no power plane almost as good as a sophisticated buried capacitance power plane?
 - Possible answer = impedance
 - Regional Vcc impedance $\sim 1\Omega$
 - Narrow line feeds impedance $\gg 1\Omega$
 - 2''' gap power plane impedance $\ll 1\Omega$
 - Hence large power mismatch in both cases!



A Power-plane design algorithm...





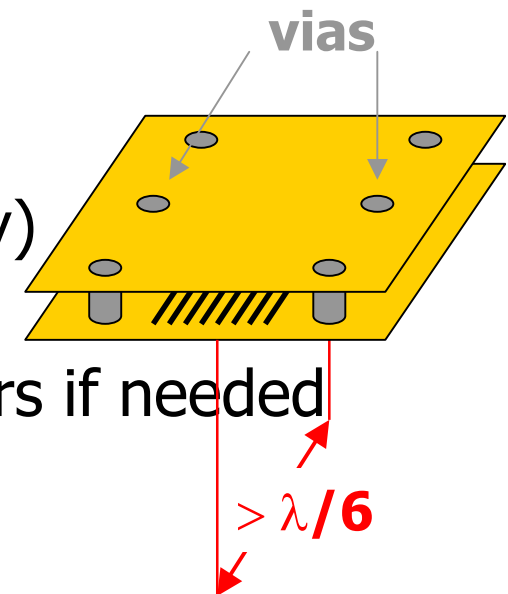
PCB Step 3 – Routing

- Use a strict routing order:

- 1st – Route power and decoupling caps
- 2nd – Route buses (stripline if necessary)
- 3rd – Route everything else
- 4th – Push non-critical tracks to PP layers if needed

- RULE3 – never let the auto router:

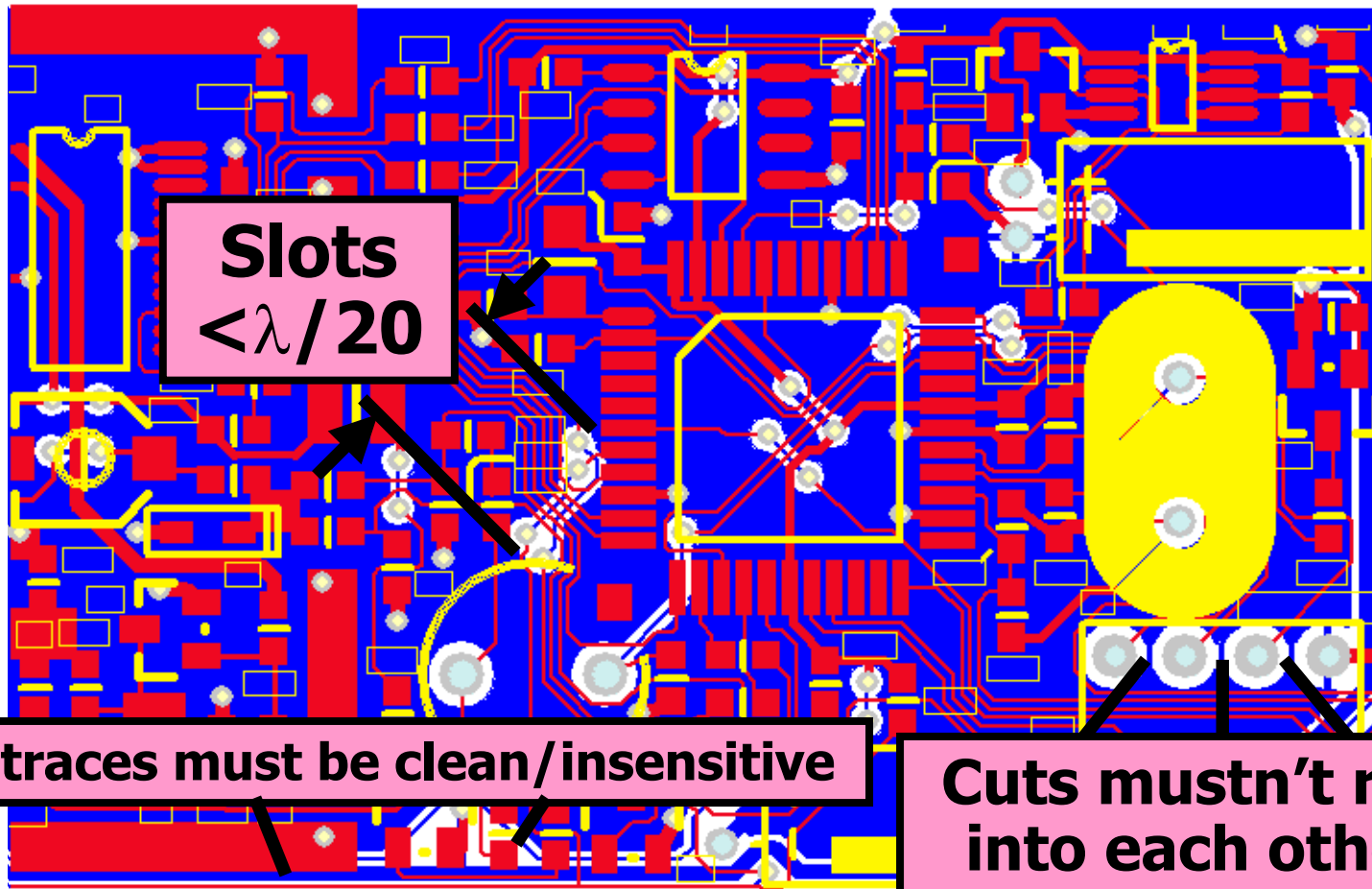
- A – Route tracks over a GND plane cut
- B – Route tracks outside a sub-section



If you can afford the time don't use the auto-router!



2 Layer PCBs & ground planes



this manual technique takes skill and time!



Routing Tips

- Be prepared to move components during the routing process.
- Minimise vias and layer changes with critical signal tracks.
- Do use circuitous (inductive) power routes
 - Form supply chains:
 - 'Clean' chain
 - 'Dirty' chain
- De-couple Vcc pins with good (small or X2Y), closely placed capacitors and small power planes (if using power planes).

A PCB design approach for best EMC (in brief)

THE END

Question from the floor

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